Rainfall Fields Monitoring Based on Satellite Microwave Down-Links and Traditional Techniques in the City of Genoa

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***Abstract*— This article compares different rainfall observing techniques available in the City of Genoa (Italy): the Ligurian regional tipping-bucket rain gauge (TBRG) network and the Monte Settepani long-range weather radar (WR) operated by the Ligurian Regional Environmental Protection Agency (ARPAL) and Smart Rainfall System (SRS), a network of microwave sensors for satellite down-links developed by the University of Genoa and Artys Srl. SRS and WR are the two indirect monitoring systems considered in this analysis. The respective performance of the rainfall measurements was assessed by comparison with the reference observations made *in situ* by seven TBRG stations selected within the boundary of the City of Genoa. The analysis was performed by considering the main precipitation events that occurred between January 2017 and December 2018. The results show an agreement between the precision of the 10-min rainfall intensity (RI) measurements, quantified by the root mean square errors (RMSEs), made by the WR and the SRS system for the seven sites. Similar performance has been also observed when comparing the peak of RI and the total rain accumulation with the reference TBRG values for the selected precipitation events. An assessment of the WR and SRS accuracy in providing the 2-D RI fields was also performed by computing the map of the RMSE differences over the City of Genoa territory. It is shown that the SRS RI maps are generally associated with a lower level of RMSE with respect to the WR in the areas that are characterized by a higher density of sensors.**

***Index Terms*— Atmosphere, electromagnetics for remote sensing, geophysical data, satellite-to-earth microwave down-links (SMLs).**

I. Introduction

GLOBALLY, the risks from extreme weather are significantly increasing, mainly because a larger number of people and assets are being exposed to this family of natural hazards [2].

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In the past 20 years, 953 natural disasters affected the European society in both human and economic terms with 88 671 casualties and more than 29 million others [32]. Recent studies show that this trend has also been confirmed in a large number of European countries [23]. In Italy, 7 275 municipalities (91% of the total) are located in areas of risk concerning floods, 16% of the national territory is exposed to high risk, and more than six million inhabitants live in high flooding risk areas [30]. In this context, the anticipation of the effects of heavy precipitation storms on partially urbanized small-medium sized river basins is a prime objective of hydrological research and environmental protection operators. The availability of systems that provide real-time precipitation monitoring is fundamental to allow environmental protection agencies to efficiently manage emergencies in order to limit damages and losses.

In the north Mediterranean region, the territories characterized by a complex orography and small natural/urban catchments (such as those of Provence-Alpes-Côte d'Azur in France or Liguria [9], north Tuscany and Campania in Italy) are often associated with high flood hazard levels. Silvestro *et al.* [29] described a typical flash flood event that occurred in the Liguria region (Italy) on November 4, 2011, caused by an intense and localized mesoscale convective system that poured very high quantities of rainfall (the Vicomorasso rain gauge measured nearly 180 mm/h and over 500 mm/6 h [9]) over a few square kilometers. As earlier highlighted by Vulpiani *et al.* [33], the effective implementation of detection and warning procedures dedicated to severe weather and related hydro-geological risks is related to the availability of responsive precipitation monitoring systems.

The regional observing systems are typically composed by a network of traditional monitoring sensors, such as tipping-bucket rain gauges (TBRGs), that perform measurements at a point-scale and long-range weather radars (WRs) that are capable to describe the spatial evolution of large rainfall cells in those areas that are not affected by the orography shielding effect [13], [34].

To achieve real-time monitoring of the time-space evolution of precipitation fields with a limited number of measuring stations, Colli *et al.* [7] proposed and tested the use of satellite-to-earth microwave down-link (SML) as a complementary technique in support of the regional TBRGs and WR monitoring networks.

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The SML solution represents an alternative to commercial microwave links (CMLs), a near ground rainfall estimation approach that exploits terrestrial microwave transmissions operated as back-haul of the cell phone network (see [4], [5], [11], [20], [22], [31], [37]). As highlighted by Chwala and Kunstmann [5], CML has a well-known limitation in adequately detecting small-scale rain events in regions characterized by sparse or without terrestrial microwave link coverage. Furthermore, the CML method requires specific agreements with the cellular companies to get access to the statistics of the microwave signal and the provision of raw data for real-time applications, such as rainfall monitoring, is often unpractical.

Differently from CML, SML can be easily installed at any location with minimal technical requirements since one sensor is composed of a low-power Internet of Things (IoT) electronic board and a common parabolic antenna used for television. Furthermore, the SML technique consists of measuring the power of the Ku-band electromagnetic field radiated by geostationary satellites, a quantity that can be measured with a high sampling interval for real-time applications and is available worldwide. On the other hand, the use of SML to retrieve rainfall measurements needs the knowledge of the melting layer as an important parameter and the actual power of the signal emitted by geostationary satellites is not known.

This recent measuring technique is supported by the field experiments made in France by Barthés and Mallet [3], Mercier *et al.* [21], in Indonesia by Indrayani *et al.* [17], and in Italy by Giannetti *et al.* [14] which highlighted a strong correlation between down-links attenuation in the Ku-band and the drop size distribution for stratiform and deep convective rain. In 2018, a comparative analysis between the rainfall intensity (RI) measured by a small set of satellite microwave down-link sensors, called Smart Rainfall System (SRS) [10], and corrected reference measurements made by dynamically calibrated TBRG [6] showed that the SRS technique can be used to update every minute the information on the average rainfall amounts measured along the link path [7]. Such aggregation time represents the current standard in rainfall monitoring as recommended by World Meteorological Organization (WMO) [36] for traditional gauges. Differently from the TBRG and WR rainfall products, the CML and SML rainfall retrieval techniques are still an object of development and lack of standard procedures enables their use as an operational tool for hydro-meteorology services. The ongoing research activities are aimed at quantifying the level of uncertainty associated with the approximations involved in such opportunistic techniques.

This article builds upon the results obtained by our previous research [7] that was focused on the evaluation of the link-averaged RI measured by single SML sensors. The present research has the objective to investigate the possibility of computing real-time rainfall maps starting from the measurements made by a capillary network of SRS microwave links sensors at the urban scale in the City of Genoa (Italy). As recommended in the conclusions of [7], the precipitation fields, based on the reflectivity figures of the polarimetric WR (Savona, Italy) of Monte Settepani and the network of TBRGs operated by Ligurian Environmental Protection Agency (ARPAL), provide validated terms of comparison for the analysis.

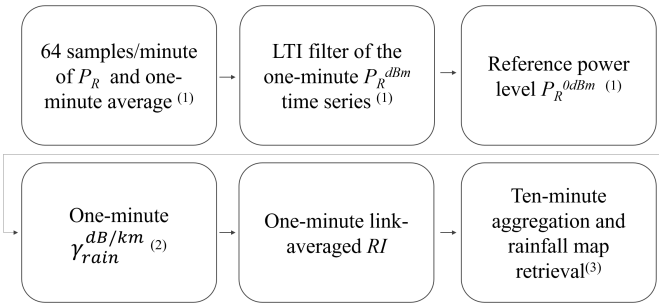


Fig. 1. SRS data processing flowchart. (1) Based on [7]. (2) Based on the melting layer retrieval approach discussed in Section III-A. (3) Rainfall field reconstruction method described in Section III-B.

The network of SRS sensors in the City of Genoa and the rainfall retrieval algorithms are described in Section II, which also provides information about the traditional rainfall measuring systems considered in the comparative analysis. A description of the method of investigation and the precipitation events that occurred in Genoa during the years 2017 and 2018 and selected for the analysis is reported in Section III. Sections IV and V include the discussion of the results and conclusive considerations on the research, respectively.

II. Description of the Measuring Systems

A. Satellite Microwave Down-Links

The satellite microwave down-link network of sensors is part of SRS, a rainfall monitoring system designed by the University of Genoa and Artys Srl for the provision of real-time RI field. SRS computes bidimensional maps that are obtained starting from RI measurements averaged along the portion of down-link that crosses the 3-D solid precipitation in the atmosphere (as illustrated in Fig. 1 of Colli *et al.* [7]). To this end, [7] illustrated the theoretical framework of SRS and the analytical description of the propagation of electromagnetic waves radiated by the satellites providing Digital Video Broadcasting (DVB-S) services.

The SRS sensors deployed in the present experiment provide measurements of the power available at the receiver P_R [dB] of the Ku-band electromagnetic field, radiated by the geostationary satellite. Following the mathematical formulation of the microwave propagation physics described in [1] and [7, eq. (5)], the rainfall specific attenuation, $\gamma_{\text{rain}}^{\text{dB/km}}$, of an SRS sensor is continuatively computed by means of the following expression:

$$\gamma_{\text{rain}}^{\text{dB/km}} = (P_R^{0\,\text{dBm}} - P_R^{\text{dBm}}) l_{\text{rain}}^{\text{km}\,-1} \tag{1}$$

where P_R^0 [dB] is the received power in the case of no-rain and $l_{\text{rain}}^{\text{km}}$ is the length of the path eventually interested by the rain.

In this article, we also adopted the method proposed in [7] to evaluate $l_{\text{rain}}^{\text{km}}$, hence

$$l_{\text{rain}}^{\text{km}} = \frac{(h_{\text{rain}} - h_{\text{ant}})}{\sin\theta} \times 0.001. \tag{2}$$

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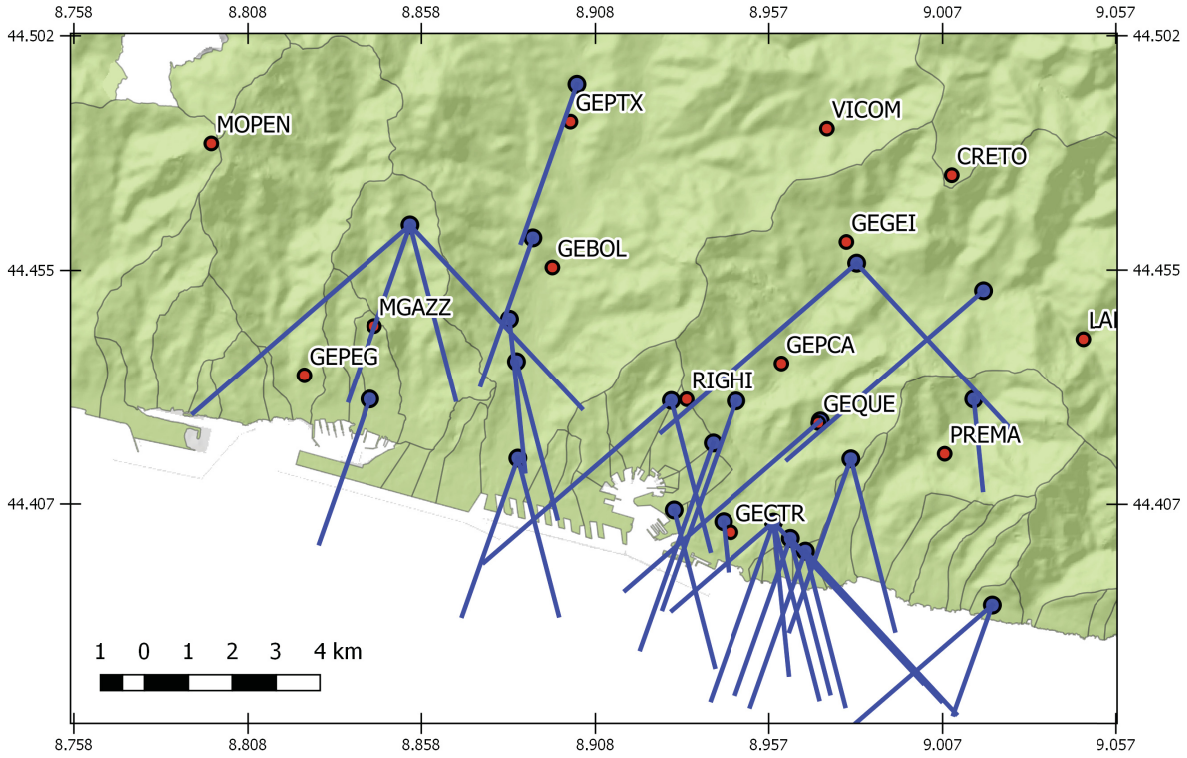


Fig. 2. Map of the horizontal projection of the microwave link paths (blue lines) associated with the SRS sensors (blue circles) functioning during the field experiment in the City of Genoa (Italy). The length of the link paths shown in the map is associated with a reference value of h_{rain} equal to 3500 m. The spatial distribution of the Liguria Region Hydro Meteorological Observatory TBRG stations is highlighted by means of red circles.

h_{rain} [m] and h_{ant} [m] are the melting layer and the antenna height, respectively, and θ [rad] is the elevation angle of the satellite microwave link as exemplified in [7, Fig. 1]. Common values of θ in the City of Genoa associated with different geostationary satellites are summarized in Table IV.

The level of the melting layer h_{rain} [m] is not known but can be estimated starting from troposphere soundings performed twice a day by the Global Climate Observing System (GCOS) network of sensors (made available by the University of Wyoming [35]) and assuming h_{rain} equal to the zero-degree isotherm. This approach has been described in [7], and the biggest shortcomings are related to the sparseness of measuring locations (the Milan Linate airport is located 120 km away from Genoa) and the long time period between measurements (12 h). For these reasons, such an estimate of h_{rain} may not be representative of the fast-convective storms occurring over the City of Genoa.

An alternative approach is adopted in this article and it consists of the vertical extrapolation of the zero-degree level starting from environmental temperature measurements made by two weather stations located at a small geographic distance but at a different altitude. In this case, it is possible to estimate h_{rain} with a very short time period (the same of the weather station measurements) and in the same area of the field experiment (the City of Genoa).

Following the International Telecommunication Union (ITU-R) P.838 international standard [18], the RI [mm h^{-1}] is formulated as a two-coefficient power-law function of the specific attenuation $\gamma_{\text{rain}}^{\text{dB/km}}$ and parametrized according to the frequency and polarization of the electromagnetic wave (by means of β and α), that is:

$$\gamma_{\text{rain}}^{\text{dB/Km}} = \beta \text{RI}^{\alpha}. \tag{3}$$

Operationally, the SRS central server receives 1-min measurements of P_R transmitted by each sensor and, in the case of attenuation due to precipitation, it updates the RI measurements every minute by assuming the reference power level P_R^0 equal to the value of P_R measured during the last no-rain time interval. The length of the path interested by the precipitation $l_{\text{rain}}^{\text{km}}$ is here simplified as the portion of the microwave down-link that is comprised between the location of the sensor and the top end of the melting layer. Considering a network of sparse satellite microwave down-links (such as in Fig. 2), it is possible to perform a bidimensional spatial reconstruction of the path-averaged RI measurements by means of a standard interpolation scheme (such as inverse distance weight (IDW) or ordinary kriging [5]) or more sophisticated approaches based on tomography techniques [8], [15]. The list of operations performed in this article to retrieve the bidimensional RI fields starting from the P_R measurements is summarized in the flowchart shown in Fig. 1.

As highlighted in [7], it is expected that the type of precipitation (e.g., convective versus stratiform and orographic) and drop size distribution have some effect on the down-link signal attenuation through the ITU-R P838 [18] parameters β and α of (3). At the moment of this article, an exhaustive analysis of the sensitivity of such parameters to the mentioned microphysical features of the precipitation is not available

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in the literature and the associated uncertainty of the RI measurements remains an open field of investigation.

The previous experimental campaign in the City of Genoa described in [7] was focused on the assessment of the SRS link-average RI performance against TBRG properly located below the satellite link. The results confirmed the capability of the SRS sensors to detect the rain/no-rain condition with the 89% of rain conditions detected by the GEDIT SRS and the 88% detected by the GEDIC SRS. The analysis of the average correlation coefficients evaluated for the GEDIT SRS sensor (which was characterized by the best position of the respective reference system below its link) is equal to 0.80 with maximum values equal to 0.89 computed during the event on January 1, 2018. Anyway, it was shown that the link-averaged RI measurements during an highly convective summer storm (July 11, 2017) were not able to describe the strong time variability of the precipitation fields, suggesting the need for performing a dedicated field experiment supported by RADAR measurements and based on the use of a capillary distribution of SRS sensors for bidimensional RI field reconstruction.

B. Description of the Traditional Rainfall Measuring Systems

The point/rain gauge data from the stations which impacting the municipality of Genoa were utilized. The data were supplied by the Liguria Region Hydro Meteorological Observatory (OMIRL), managed by ARPAL, which counts all over the Liguria region from over 200 rainfall stations. Each station has a sampling time ranging from 5 to 30 min and is part of the Italian Operation Network which has in excess of 1500 stations, making it one of the densest networks in the world [29].

Table V lists a set of stations that have been considered to perform the point-scale comparison between SRS and TBRG measurements and that are located in the proximity of the satellite down-links. Differently, the whole set of OMIRL stations of the City of Genoa has been considered to compute the reference rainfall fields for the spatial analysis shown in Figs. 6 and 9.

The radar data used were provided by a C-band polarimetric radar located at approximately 1400 m above mean sea level on Monte Settepani in the province of Savona. The lon/lat coordinates of the radar are 8.197466° and 44.245894° (WGS84/EPSG4326 reference system), respectively.

The radar is used by the Meteorological Weather Services of the regions of Liguria and Piedmont, and also by the Italian Department of Civil Protection. The RADAR rain-rate estimation is based on a flowchart decision tree that allows the use of the best rain-rate retrieval algorithm, depending on the value of polarimetric variables [27]. In particular, the methodology takes into account differential phase filtering, attenuation correction, and the application of a multiparametric algorithm for rainfall estimation, which is derived from the approach proposed by Colorado State University (see [24]). The algorithm flowchart combines single tests for deciding which formulation for rain intensity estimation is the most suitable. The aim is to use alternatively the optimal measured variables for calculating the precipitation; in this way, each moment is used in the range where its estimates can be considered more reliable, and the rain-rate formulas give physically reliable results [27].

TABLE I

CHARACTERISTICS OF THE MONTE SETTEPANI RADAR, SAVONA, LIGURIA (MORE DETAILS ARE GIVEN IN [24] AND [25])

Radar model	GPM250C (Selex-Gematronik)
Radar elevation	1,400 m
Beam width	1°
Maximum range	134 km
PRF (Hz)	1040
Temporal resolution	5 min.
Pulse length	0.5 μs
Number of elevations	11
Measured variables	Z_H, Z_{DR}, ϕ_{DP}, ρ_{HV}, V_r

An algorithm for clutter elimination is applied in real time to raw radar data before calculating rainfall [26]. Rain rate is then calculated by considering data from the lowest unshielded elevation, identified by an analysis of visibility based on a digital elevation model.

Precipitation maps are produced with a spatial resolution of 1×1 km and a temporal resolution of 10 min. For this article, the processed radar data were provided by ARPAL. The main characteristics of the Monte Settepani radar are shown in Table I.

III. METHOD OF INVESTIGATION AND DATA SET

A. Melting Layer Estimation

As discussed in Section II, the computation of the SRS path-averaged RI requires the evaluation of the altitude of the rain cell [h_{rain} in (2)]. In this analysis, h_{rain} is represented by the level of the melting layer (hereinafter m.l.), a variable that is updated on an hourly basis by measuring the environmental temperatures at two different meteorological stations of OMIRL (identified in Table II with the STILA and FFRES identifiers) and characterized by a different altitude notwithstanding a short horizontal distance (Fig. 3). The horizontal distance between STILA and FFRES stations is equal to 3.3 km while the vertical distance is equal to 617 m. Thanks to such geographical configuration, the vertical temperature gradient measured by the two stations can be used to estimate the zero degree level h_{rain} by accepting the approximation of a linear vertical temperature profile.

Notwithstanding the level of uncertainty ascribed to the assumption of a linear vertical temperature profile, this method has two major advantages with respect to the use of upper air soundings described in Colli *et al.* (2018) [7]. The first is the possibility to update h_{rain} with a shorter time interval (in our study equal to 1 h) allowing the description of the quick time evolution of precipitation storms in Genoa. A further benefit is due to the possibility to evaluate h_{rain} in the geographic area of rainfall monitoring.

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TABLE II

OMIRL METEOROLOGICAL STATIONS CONSIDERED TO ESTIMATE THE REFERENCE ENVIRONMENTAL TEMPERATURE MEASUREMENTS

Station ID	Site name	Latitude *WGS84/ EPSG4326*	Longitude *WGS84/ EPSG4326*	Altitude *m*
STILA	S. Ilario	44.384°	9.061°	174
FFRES	Fontana Fresca	44.403°	9.093°	791

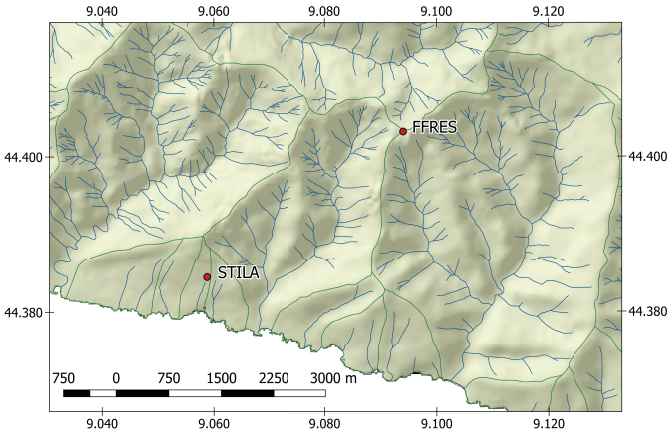


Fig. 3. Map of the OMIRL meteorological stations used to provide the reference temperature measurements (red circles) during the field experiment in the City of Genoa (Italy).

In order to assess the level of uncertainty associated with the use of a specific m.l. retrieval method, the attenuation measurements made by the GEDIC SRS sensor during the previous Colli *et al.* [7] experiment was used to re-compute link-averaged rainfall estimates by adopting both approaches. Reference measurements were provided by two dynamically calibrated TBRGs located under the GEDIC link by processing the TBRG raw data with an advanced sampling procedure [7]. The upper air soundings measurements were updated every 12 h while comparison between the SRS and rain gauge precipitation measurements was based on 10-min rainfall accumulation data set; more details about the data set are reported in [7]. The main statistics of the relative error between SRS and rain gauge 10-min accumulation are reported in Table III and classified according to the RI interval. The results show that the two m.l. evaluation approaches led to errors characterized by a similar order of magnitude. The evaluation based on the use of weather stations slightly improves the accuracy of the rainfall measurements, represented by the averaged error avg(e_{SRS}), in case of medium and intense RI. On the other hand, the precision of the SRS measurements, represented by the standard deviation of the error st.dev(e_{SRS}) in Table III, is significantly increased when the upper air soundings method is adopted at the lowest RI class ($0\ \text{mm}\cdot\text{h}^{-1} < \text{RI} <= 12\ \text{mm}\cdot\text{h}^{-1}$). At higher RI classes, such behavior is less relevant with closer st.dev(e_{SRS}) figures among the two methods.

In the light of such results and the availability of hourly ground measurements performed close to the testbed area, the m.l. assessment used in the present analysis is based on the use of weather station measurements and the assumption of an evaluated linear vertical temperature profile.

TABLE III

AVERAGE AND STANDARD DEVIATION OF THE RELATIVE ERRORS OF RI MEASUREMENTS MADE BY THE GEDIC SRS SENSOR [7] AND PROCESSED BY ADOPTING THE WEATHER STATION (W.S.) AND THE SOUNDINGS MELTING LAYER RETRIEVAL METHODS

RI class	W.S. method		Soundings method	
	avg(e_{SRS})	st.dev(e_{SRS})	avg(e_{SRS})	st.dev(e_{SRS})
mm h^{-1}	-	-	-	-
0-12	1.24	1.29	1.07	1.13
12-24	0.12	0.39	0.07	0.36
24-36	-0.18	0.29	-0.24	0.28
36-48	-0.38	0.20	-0.42	0.20
48-60	-0.49	0.16	-0.52	0.16
60-72	-0.50	0.14	-0.53	0.12
72-84	-0.67	0.13	-0.71	0.11

TABLE IV

LIST OF SATELLITE MICROWAVE DOWN-LINKS COMPOSING THE SRS NETWORK OF SENSORS IN THE CITY OF GENOA (ITALY)

Satellite name	Azimuth φ [deg]	Elevation θ [deg]	N. sensors
Astra 1L	165.5° N	37.8°	9
Eutelsat 5 West	199.5° N	36.9°	12
Hispasat 30W-5	229.1° N	25.9°	7
Hot Bird 13C	174.2° N	38.6°	4
Turksat 4A	137.1° N	29.2°	5

TABLE V

OMIRL RAIN GAUGES CONSIDERED FOR PROVIDING REFERENCE MEASUREMENTS IN THE CITY OF GENOA

Station ID	Site name	Latitude *WGS84/ EPSG4326*	Longitude *WGS84/ EPSG4326*
GECTR	ARPAL center	44.4004°	8.9459°
MGAZZ	Monte Gazzo	44.443288°	8.844321°
GEPTX	Pontedecimo	44.484993°	8.900546°
GEQUE	Quezzi	44.423373°	8.971616°
GEPCA	Staglieno	44.435617°	8.961021°
PREMA	Premanico	44.417094°	9.007775°
RIGHI	Righi	44.428256°	8.933783°

B. Rainfall Map Retrieval and SRS Sensors Set-Up

The Monte Settepani WR RI maps are provided on a rectangular grid uniformly discretized in 400 intervals in the lon/lat directions with a spatial resolution equal to 1 km. The lon/lat coordinates of the bottom-left corner are 5.792018° and 42.4047187° (WGS84/EPSG4326 reference system), respectively.

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TABLE VI

CHARACTERISTICS OF THE RAINFALL EVENTS SELECTED FOR THE ANALYSIS MEASURED BY THE GECTR, STILA, AND FFRES STATIONS

ID	Time start	Time end	max(RI)	h_{TOT}	avg(T_{STILA})	avg(T_{FFRES})	m. l. level
	UTC	*UTC*	*mmh^{-1}*	*mm*	*°C*	*°C*	*m*
Ev01	05/02/2017 05:00	12:00:00	9.4	24.9	10.3	6.0	1652
Ev02	09/09/2017 13:00	18:00:00	39.6	17.4	18.3	14.3	2997
Ev03	05/11/2017 04:00	10:00:00	33.6	17.6	13.0	8.6	1997
Ev04	27/12/2017 01:00	05:00:00	24.0	18.2	9.1	5.0	1543
Ev05	01/01/2018 00:00	10:00:00	10.8	23	9.7	4.5	1325
Ev06	15/03/2018 07:00	19:00:00	14.4	35.2	11.1	6.5	1663
Ev07	17/04/2018 12:00	15:00:00	34.8	12.8	17.0	15.9	2789
Ev08	09/05/2018 02:00	08:00:00	14.4	10.8	16.6	11.6	2141
Ev09	28/06/2018 00:00	04:00:00	25.2	14.4	17.3	14.6	4127
Ev10	28/06/2018 11:00	16:00:00	36.0	12.2	19.4	15.3	3093
Ev11	04/07/2018 00:00	04:00:00	49.2	14.2	19.4	17.0	5161
Ev12	12/07/2018 06:00	12:00:00	15.6	6.8	22.2	16.7	2664
Ev13	14/08/2018 08:00	13:00:00	82.8	44.6	21.7	16.9	2963
Ev14	11/10/2018 00:00	09:00:00	10.8	13.6	17.4	13.5	2927
Ev15	27/10/2018 11:00	24:00:00	60.0	88.2	16.2	11.6	2347
Ev16	10/11/2018 01:00	11:00:00	32.4	59.2	14.2	9.7	2121

TABLE VII

CONTINGENCY MATRIX OF THE 10-MIN AVERAGED RI MEASURED BY SRS AT TBRG THE STATION LOCATIONS. THE "TRUE" GROUP REPRESENTS AN AGREEMENT CONDITION BETWEEN THE MEASUREMENTS MADE BY SRS AND THE TBRG SYSTEM WHILE THE "FALSE" CONDITION REPRESENTS THE OPPOSITE CASE. THE SIZE OF THE DATA SET FOR EACH INTENSITY CLASS AND LOCATION IS SPECIFIED IN THE SIZE COLUMN. THE RI CLASS LIMITS ARE BASED ON THE RI TBRG MEASUREMENTS

location	0 mm h^{-1}			> 0 mm h^{-1}			0-12 mm h^{-1}			12-24 mm h^{-1}			>24mm h^{-1}		
	True	False	Size	True	False	Size	True	False	Size	True	False	Size	True	False	Size
CFUNZ	80.4%	19.6%	291	92.5%	7.5%	319	33.5%	66.5%	275	40.7%	59.3%	27	58.8%	41.2%	17
PREMA	88.7%	11.3%	256	82.5%	17.5%	354	55.3%	44.7%	304	51.6%	48.4%	31	73.7%	26.3%	19
GEPTX	88.2%	11.8%	305	83.6%	16.4%	305	53.7%	46.3%	285	80.0%	20.0%	15	40.0%	60.0%	5
RIGHI	90.4%	9.6%	281	67.5%	32.5%	329	46.4%	53.6%	291	7.7%	92.3%	26	41.7%	58.3%	12
GEPCA	88.0%	12.0%	276	75.7%	24.3%	334	54.7%	45.3%	300	25.0%	75.0%	20	42.9%	57.1%	14
GEQUE	90.6%	9.4%	278	81.0%	19.0%	332	55.9%	44.1%	286	31.0%	69.0%	29	11.8%	88.2%	17
MGAZZ	93.4%	6.6%	334	51.1%	48.9%	276	24.9%	75.1%	257	53.8%	46.2%	13	83.3%	16.7%	6
tot	88.5%	11.5%	2021	76.3%	23.7%	2249	46.3%	53.7%	1998	41.4%	58.6%	161	50.3%	49.7%	90

In this article, the bidimensional rainfall maps of SRS are computed by discretizing the spatial domain with a structured rectangular mesh that is consistent with the WR discretization to ensure the comparability of the results. The link-averaged measurements of the 10-min RI were used to compute the values at the barycenter of each grid cell by using the common IDW interpolation scheme with a Shepard's power parameter [25] equal to 1.

The extent of the SRS precipitation field of the Genoa area is defined by the geographical limits coordinates, respectively, equal to 8.8° and 9.05° in the longitudinal direction and 44.35° and 44.5° in latitudinal direction (WGS84/EPSG4326 reference system).

The network of SRS sensors operating during the field investigation in the City of Genoa was composed of 37 microwave down-links installed in 21 different locations (Fig. 2). A choice of five different satellites (Astra 1L, Eutelsat 5 West, Hispasat 30W-5, Hot Bird 13C, and Turksat 4A) has been considered to achieve the largest territory coverage with a minimal number of sensors. Table IV reports on the number of SRS sensors grouped by satellite and includes information about the down-link azimuth orientation and elevation angle of the microwave links.

The interface circuit between the LNB (Low Noise Block) and the onboard processing unit of SRS provides the power supply and the 22-kHz tone to select the proper sub-band,

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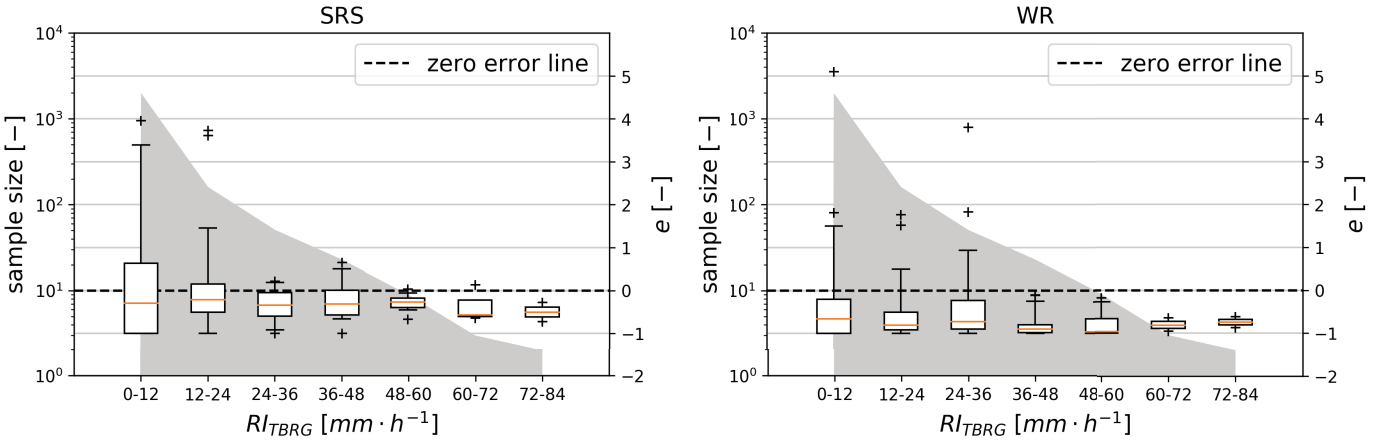


Fig. 4. Relative error e [−] of 10-min RI measurements made by (Left) SRS and (Right) WR for different RI_{TBRG} classes [mm h^{-1}] measured by the reference TBRG.

TABLE VIII

CONTINGENCY MATRIX OF THE 10-MIN AVERAGED RI MEASURED BY WR AT TBRG THE STATION LOCATIONS. THE "TRUE" GROUP REPRESENTS AN AGREEMENT CONDITION BETWEEN THE MEASUREMENTS MADE BY WR AND THE TBRG SYSTEM WHILE THE "FALSE" CONDITION REPRESENTS THE OPPOSITE CASE. THE SIZE OF THE DATA SET FOR EACH INTENSITY CLASS AND LOCATION IS SPECIFIED IN THE SIZE COLUMN. THE RI CLASS LIMITS ARE BASED ON THE RI TBRG MEASUREMENTS

location	0 mm h^{-1}			> 0 mm h^{-1}			0-12 mm h^{-1}			12-24 mm h^{-1}			>24mm h^{-1}		
	True	False	Size	True	False	Size	True	False	Size	True	False	Size	True	False	Size
GECTR	92.4%	7.6%	291	62.4%	37.6%	319	42.2%	57.8%	275	11.1%	88.9%	27	29.4%	70.6%	17
PREMA	89.5%	10.5%	256	75.1%	24.9%	354	53.3%	46.7%	304	19.4%	80.6%	31	42.1%	57.9%	19
GEPTX	90.8%	9.2%	305	74.8%	25.2%	305	57.5%	42.5%	285	20.0%	80.0%	15	0.0%	100.0%	5
RIGHI	94.0%	6.0%	281	81.2%	18.8%	329	62.5%	37.5%	291	11.5%	88.5%	26	25.0%	75.0%	12
GEPCA	92.4%	7.6%	276	84.1%	15.9%	334	63.0%	37.0%	300	10.0%	90.0%	20	7.1%	92.9%	14
GEQUE	88.8%	11.2%	278	78.0%	22.0%	332	58.0%	42.0%	286	3.4%	96.6%	29	11.8%	88.2%	17
MGAZZ	93.4%	6.6%	334	80.1%	19.9%	276	58.8%	41.2%	257	7.7%	92.3%	13	16.7%	83.3%	6
tot	91.6%	8.4%	2021	76.5%	23.5%	2249	56.5%	43.5%	1998	11.9%	88.1%	161	18.9%	81.1%	90

the lower Ku-band (10.75–11.7 GHz) and the higher band (11.7–12.75 GHz), and polarization. To maximize the power of the signal received by the SRS antennas, the LNB of the larger part of sensors listed in Table IV have been set to operate in the lower Ku sub-band with vertical polarization while the SRSs aligned with Turksat 4A operate in higher sub-band and vertical polarization. The large transponder population in that configuration determined this choice. Accordingly, the values of the coefficients of the ITU rain attenuation model [18] adopted in this article and expressed by (3) have been set equal to $\alpha = 1.264$ and $\beta = 0.009$.

Originally, the SRS rainfall products are computed with a 1-min refreshing time. With the aim of comparing the SRS precipitation fields with rainfall maps obtained from the Monte Settepani WR, the analysis is performed by aggregating the SRS data to 10-min time, i.e., the same interval for the WR products.

C. Assessing the Rainfall Map Performance

A first assessment of the accuracy of the SRS precipitation fields is performed by considering the TBRG measurements as the reference since they constitute the best representation of the actual RI. Under this assumption, it is possible to compute the values of the SRS RI at a selection of given TBRG locations listed in Table V and to evaluate the level of agreement with the corresponding tipping bucket observations. The same assessment is replicated starting from the WR precipitation fields. In this manner, the performance of the SRS and WR RI measurements are quantified and compared in terms of contingency tables and relative errors in Section IV.

Referring to Fig. 2, it is worth noting that the TBRG locations identified by the GECTR, GEPTX, GEQUE, and MGAZZ identifiers are conveniently located under an SRS down-link while the GEPCA, PREMA, and RIGHI locations do not exactly lie below a down-link. This different configuration of TBRG locations has been chosen in order to detect possible trends in the results that may be ascribed to the spatial interpolation technique adopted by SRS rather than the accuracy of the down-link sensors.

The relative error e_{rel} (−) of a RI measurement is defined as follows:

$$e_{\text{rel}} = \frac{(\text{RI}_{\text{meas}} - \text{RI}_{\text{ref}})}{\text{RI}_{\text{ref}}} \tag{4}$$

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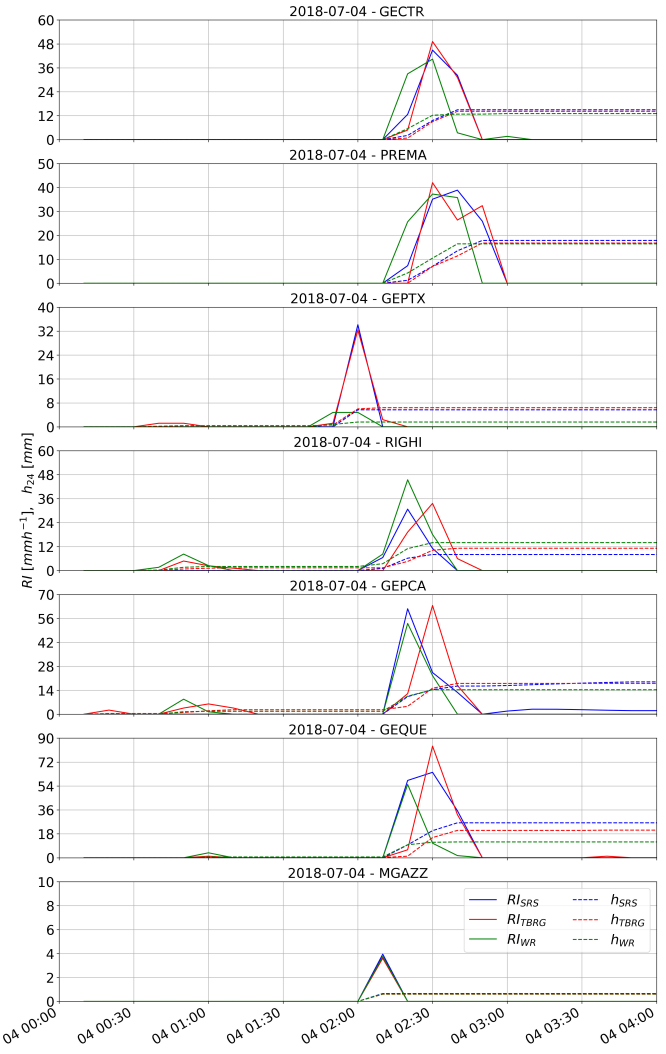


Fig. 5. Time series of the 10-min RI [mm h^{-1}] and water accumulation h [mm] measured by the SRS, TBRG, and the WR sensors (respectively represented by blue, red, and green lines) at the GECTR, PREMA, GEPTX, RIGHI, GEPCA, GEQUE, and MGAZZ locations (defined in Table V) during the Ev11 convective precipitation event.

where RI_{meas} is a RI measurement performed by SRS or the WR, and RI_{ref} is the reference value measured by the TBRG at the same location. The use of e_{rel} allows computing a statistical assessment of the system performance and representing the results in the form of box plots grouped for different reference RI classes. Specific information about the performance of the measuring systems under evaluation is included in Section IV by representing the difference between the peak of 10-min RI and the total cumulated rainfall recorded during each precipitation event considered and each location in the analysis (and, respectively, listed in Tables V and VI). Overall assessments of the relative errors, the peak RI, and the cumulated rainfall are also provided in terms of root mean square errors (RMSEs).

The analysis is concluded by representing the spatial distribution of the difference between the WR and SRS RMSEs where negative values represent areas where the overall accuracy of RI maps is higher basing on WR rather than SRS and negative values represent the opposite case.

D. Data Set

The analysis has been carried out considering a set of precipitation events that occurred between January 2017 and December 2018 in the City of Genoa.

An overview of the total cumulated rainfall h_{TOT} [mm] and the maximum RI max(RI) [mm · h^{-1}] measured by the GECTR station is reported in Table VI for each precipitation event. Furthermore, the list includes the average environmental temperatures avg(T) [°C] measured by the FFRES and STILA stations and considered to compute the altitude of the melting layer. The data set is composed of events characterized by $h_{TOT} > 6$ mm and occurred when all the measuring systems considered in this analysis were fully operational.

In the greater part of the selected cases, precipitation was associated with large-scale disturbances or frontal systems coming from the west or southwest. In some of these cases (namely, Ev02, Ev08, Ev11, Ev13, Ev15), convective systems with enhanced RI and lightning activity were also observed, especially during the latter two events. A different category of events, typically during late spring and summer, were more purely convective ones and driven either by diurnal heating and orographic lifting (Ev07, Ev10) or by the passage of small-scale upper level disturbances destabilizing the atmosphere during the night (Ev09, Ev12). In the remaining cases (Ev04, Ev14, Ev16), precipitation originated over the Ligurian Sea and was advected inland within a moist, prefrontal southerly or southwesterly flow. In particular, in the last considered case, a quasi-stationary convective system, persisting several hours, was triggered over the sea by a low-level convergence between a moist southeasterly flow and a cooler, drier northerly flow from the Po Valley.

The SRS sensor measurements are sent to the Artys Srl server every minute by using GSM cellular telecommunication or local Internet connection if available at the test sites. In a similar manner, the TBRG measurements are continuously collected by the ARPAL server via radio telecommunication and made available with a 5-min updating time. The WR post-processed products were collected by ARPAL, and the RI maps are available with a 10-min updating time. In order to compare the three different rainfall products, we performed the following operations.

1) *TBRG:* The 5-min rain accumulation measurements were aggregated to 10-min intervals (the same of the RADAR data set) and converted in terms of 10-min averaged RI.
2) *SRS:* The ten 1-min RI samples within each 10-min RADAR interval were averaged to produce 10-min averaged RI series.

To support the analysis, a data quality check has been performed so as to detect and discard time intervals where the SRS, WR, and TBRG measurements were not simultaneously available.

IV. Results and Discussion

A first indication of the level of agreement between the 10-min averaged RI estimates provided by SRS maps and ground-based TBRG measurements is provided in the form of a contingency matrix in Table VII. This representation

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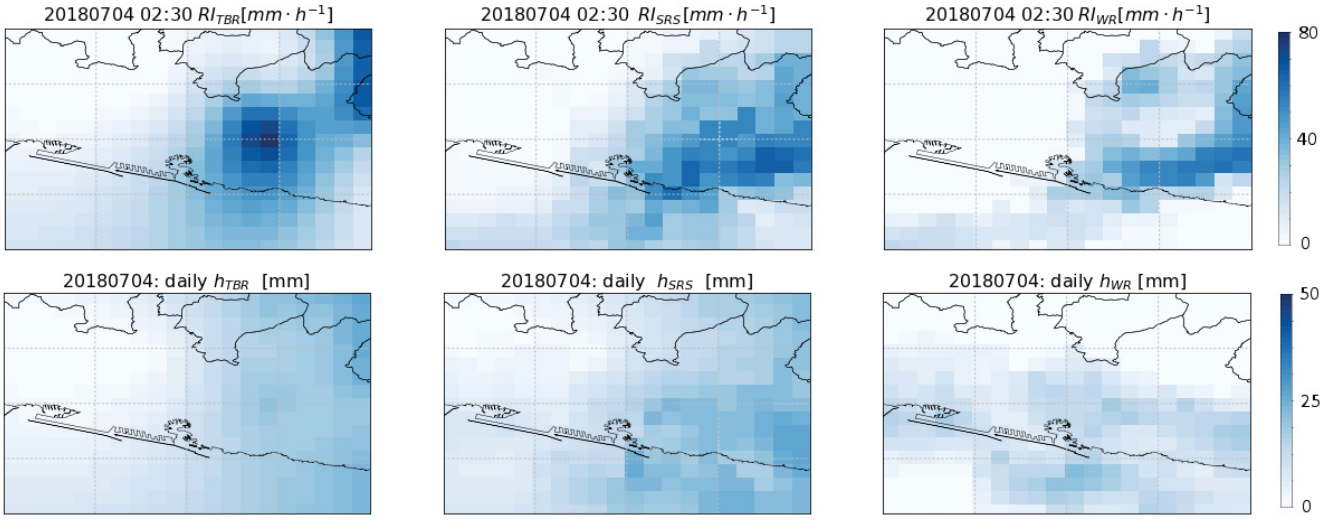


Fig. 6. (Top) Bidimensional representation of the RI [mm h^{-1}] fields measured by, from left to right, the TBRG, SRS, and the WR for the 10-min interval characterized by the peak of RI_{TBRG} during the Ev11 convective precipitation event. (Bottom) Bidimensional daily rainfall accumulations fields h [mm] measured by, from left to right, the TBRG, SRS, and the WR systems during Ev11.

allows expressing the percentage of success of the SRS map in recognizing the actual RI class occurring at the TBRG locations. The same approach is adopted to evaluate the WR performance in Table VIII. The contingency matrixes of RI showed that the level of agreement between 10-min averaged RI values provided by SRS maps and ground-based TBRG measurements is comparable with the results achieved by WR products.

Some differences can be noted when RI < 12 mm $\cdot$ h^{-1}, where the total percentage of success assumes lower values with the SRS data set, equal to 88.5% and 46.3%, respectively, for the 0-mm $\cdot$ h^{-1} and the 0–12-mm $\cdot$ h^{-1} classes, in comparison with the WR figures, equal to 91.6% and 56.5%, respectively, for the 0-mm $\cdot$ h^{-1} and the 0–12-mm $\cdot$ h^{-1} classes.

When the RI class is above 12 mm$\cdot$h^{-1}, the total percentage of agreement between SRS and TBRG assumes higher values than the comparison between WR and TBRG. For example, the $>$24-mm $\cdot$ h^{-1} class shows a total agreement percentage equal to 50.3% for SRS against 18.9% for WR. It must be noted that the percentage of agreement between the 10-min RI values measured by both the SRS and WR and the TBRG point-scale observations assumes low values above the 24 mm $\cdot$ h^{-1} limit, confirming the general assumption that these areal remote sensing techniques can provide complementary information with respect to the measurements performed by TBRGs. The former should be used to timely monitor the space evolution of the precipitation fields while the latter permits accurate measurements of the RI magnitude at fixed locations. These results also show a relevant number of "missing hits" situations where the reference RI > 0 mm$\cdot$h^{-1} but the SRS and WR systems fail in detecting precipitation at the TBRG location (disagreement percentage equal to, respectively, 23.7% and 23.5%).

A more detailed picture of the performance obtained by SRS and WR RI measurements is described in terms of the statistic distribution of the relative error e [as of (4)]. The box plots of Fig. 4 illustrate the different percentiles of e as a function of the reference RI RI_{TBRG} measured by the whole set of TBRG listed in Table V and precipitation events listed in Table VI. Fig. 4(a) reports about the results obtained by SRS while Fig. 4(b) shows the results of the WR data set.

It is shown that the measurements of SRS are characterized by larger boxes and whiskers than those derived by WR at low intensities (below $RI_{TBRG} = 24$ mm $\cdot$ h^{-1}). At higher rainfall intensities, the width of the boxplots shows a similar behavior between the two systems. Larger boxplots are generally associated with a lower precision of the measuring system under evaluation.

For each box, the mean values of the error avg(e) are represented by an orange line contained inside the rectangle. The magnitudes of the mean error (|avg(e)|) that are close to 0 represent high trueness of the measuring system under evaluation. The comparative representation proposed in Fig. 4 shows that SRS is characterized by |avg(e)| values that are closer to 0 than the ones of the WR figures. In general, it is possible to conclude that the 10-min RI_{SRS} values interpolated at the TBRG locations of Table V show higher trueness than the WR measurements and a lower precision during light rainfall intensities.

As an example, Fig. 5 shows the SRS, TBRG, and WRR time series of the 10-min RI (mm $\cdot$ h^{-1}) and the cumulative rainfall height h (mm) for a typical convective rainfall storm in Liguria (Italy) measured at the seven TBRG locations. The different panels of Fig. 5 demonstrate the high variability of the peak of the reference RI^p and the 24-h rainfall accumulation among the different sites due to the small spatial extent of intensive convective cells [18] at such distances ($RI^p = 4.0$ mm$\cdot$h^{-1} at MGAZZ while $RI^p = 62.6$ mm$\cdot$h^{-1} at GEPCA). Fig. 5 highlights that, under such conditions, both the SRS and the WR measurements show significant deviations with respect to the reference TBRG time series. The same case study has been considered in the bidimensional rainfall field representations proposed in Fig. 6. The panels located at the top of Fig. 6 illustrate, from left to right, the RI fields based on

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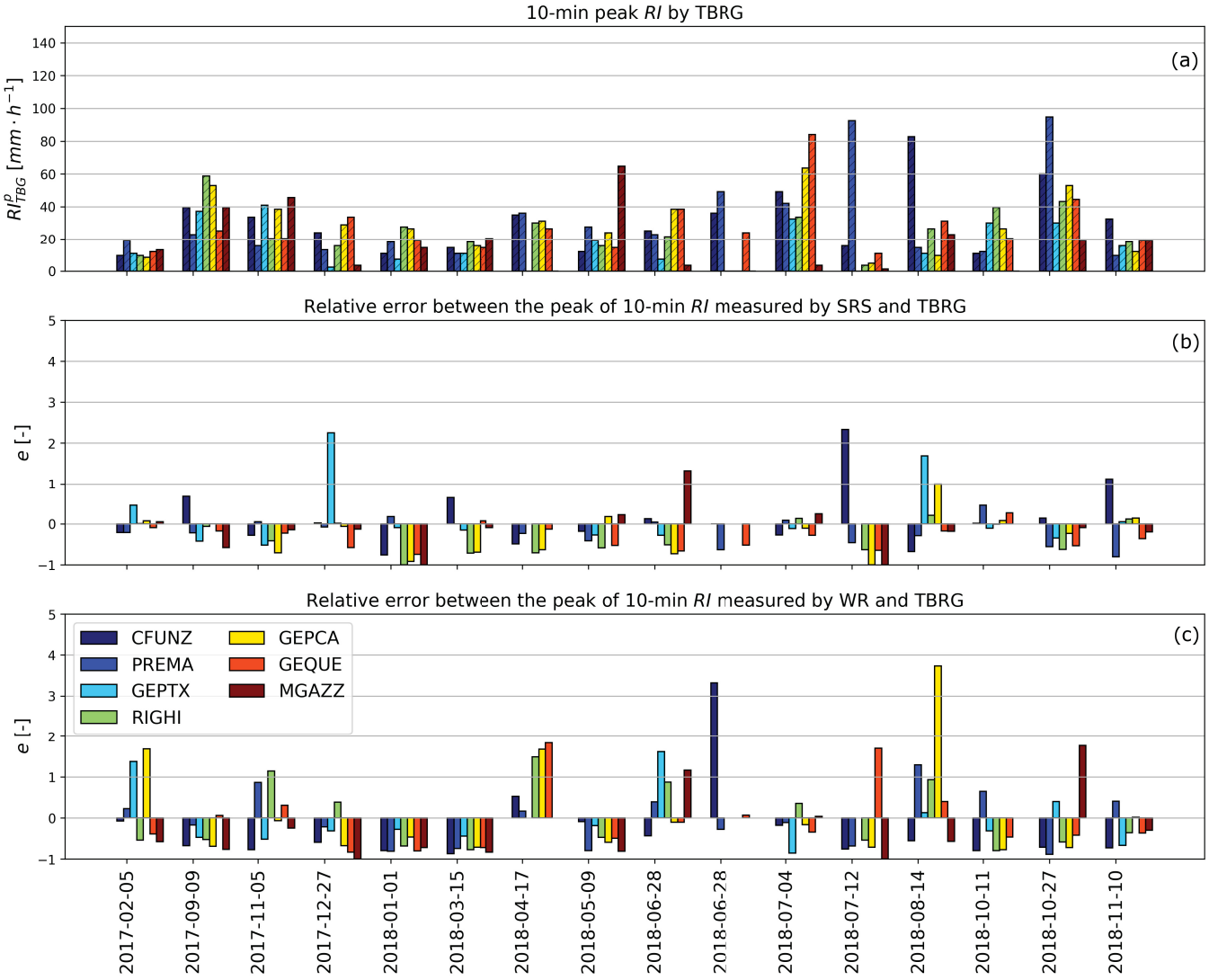


Fig. 7. (a) Reference RI^p_{TBRG} [mm h^{-1}] measured by the TBRGs during the field campaign (b) Relative error e [−] between the peak of RI measured by SRS RI^P_{SRS} [mm h^{-1}] and RI^p_{TBRG} for the GECTR, PREMA, GEPTX, RIGHI, GEPCA, GEQUE, and MGAZZ locations (defined in Table V) during any precipitation event considered in the analysis. (c) Relative error of RI^P measured by WR with respect to the TBRG data set.

TABLE IX

ABSOLUTE AND NORMALIZED RMSE (RMSE AND RRMSE) OF THE RI PEAK MEASURED BY SRS AND WR (RI^PSRS AND RI^PWR) AT THE TBRG LOCATIONS (SEE TABLE V)

Station ID	RMSE(RI^P_{SRS})	RMSE(RI^P_{WR})	rRMSE(RI^P_{SRS})	rRMSE(RI^P_{WR})
	mmh^{-1}	mmh^{-1}	-	-
GECTR	21.3	36.5	1.05	1.79
MGAZZ	9.9	18.8	0.52	0.99
GEPTX	8.9	11.9	0.63	0.84
GEQUE	13.1	18.4	0.73	1.02
GEPCA	13.6	24.2	0.73	1.31
PREMA	19.2	28.2	0.71	1.05
RIGHI	12.4	20.8	0.82	1.37
Mean	14.1	22.7	0.74	1.20

the TBRG, SRS, and the WR measurements during the 10-min interval of the peak of RI measured by the GECTR, PREMA, GEPCA, RIGHI, and GEQUE stations (Fig. 5). The bottom part of the figure includes a similar representation of the daily rainfall accumulations fields measured by the three systems. A visual comparison of the RI fields shows that the SRS and the

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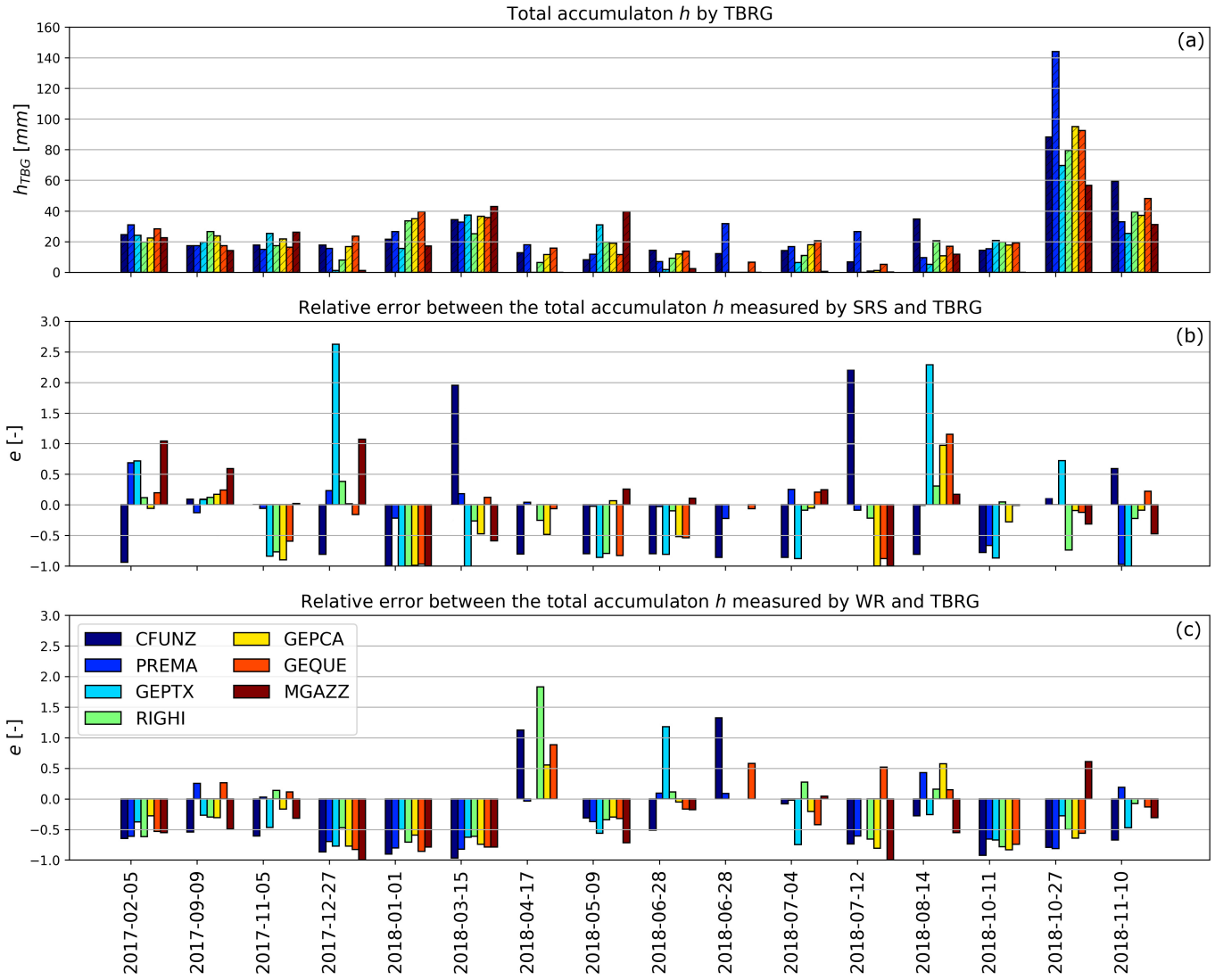


Fig. 8. (a) Reference rain total accumulation h_{TBRG} [mm] measured by the TBRGs during the field campaign. (b) Relative error e [−] between the rain total accumulation measured by SRS h_{SRS} [mm] with respect to h_{TBRG} for the GECTR, PREMA, GEPTX, RIGHI, GEPCA, GEQUE, and MGAZZ locations (defined in Table V) during any precipitation event considered in the analysis. (c) Relative error of h_{WR}.

WR systems depict different patterns of the precipitation while the spatial extent of the areas characterized by high intensity is identified by both the systems. The deviations between the SRS and WR total accumulation fields are evident as well. The spatial distribution of the TBRGs reported in Fig. 2 shows the existence of large portions of territory that are not covered by a rain gauge station; for this reason, the direct comparison between the 2-D interpolation of the TBRGs measurements and the SRS, or the WR, spatial reconstruction at small spatial scales can be hardly used to provide quantitative assessment.

The comparison between the SRS and WR performance at the rain gauges locations is indeed provided in the form of histogram plots in Figs. 7 and 8. The bar's color code identifies the TBRG location, and the histograms are grouped for precipitation events. The top panel of Fig. 7 illustrates the peak of RI measured by the TBRG ($\mathrm{RI}^{p}_{\mathrm{TBRG}}$), the central panel shows the relative error between the SRS measurements, and the bottom panel shows the relative error performed by the WR data set. Similarly, the top panel of Fig. 8 illustrates the total rain accumulation at the event scale measured by the TBRG (h_{TBRG}), the central panel shows the relative error of the SRS measurements, and the bottom panel shows the relative error of the WR measurements.

These results are also summarized over the whole precipitation events' data sets in Tables IX and X that, respectively, report about the RMSE of RI^{p} and h measurements made by SRS and WR both in absolute terms (RMSE) and normalized with the standard deviation of the reference measurements (rRMSE).

When the SRS data set is analyzed, the peak of RI assumes lower deviations with respect to the TBRG observations, demonstrating a better behavior of SRS with respect to WR in depicting high-intensity figures (Fig. 7). This behavior is confirmed by the RMSE values of RI^{p} (Table IX) which are consistently lower for each TBRG location if SRS is adopted. Consequently, averaging the rRMSE(RI^{p}) values among the different sites yields values equal to 0.74 in the case of SRS and 1.20 when WRs are considered. However, some caution

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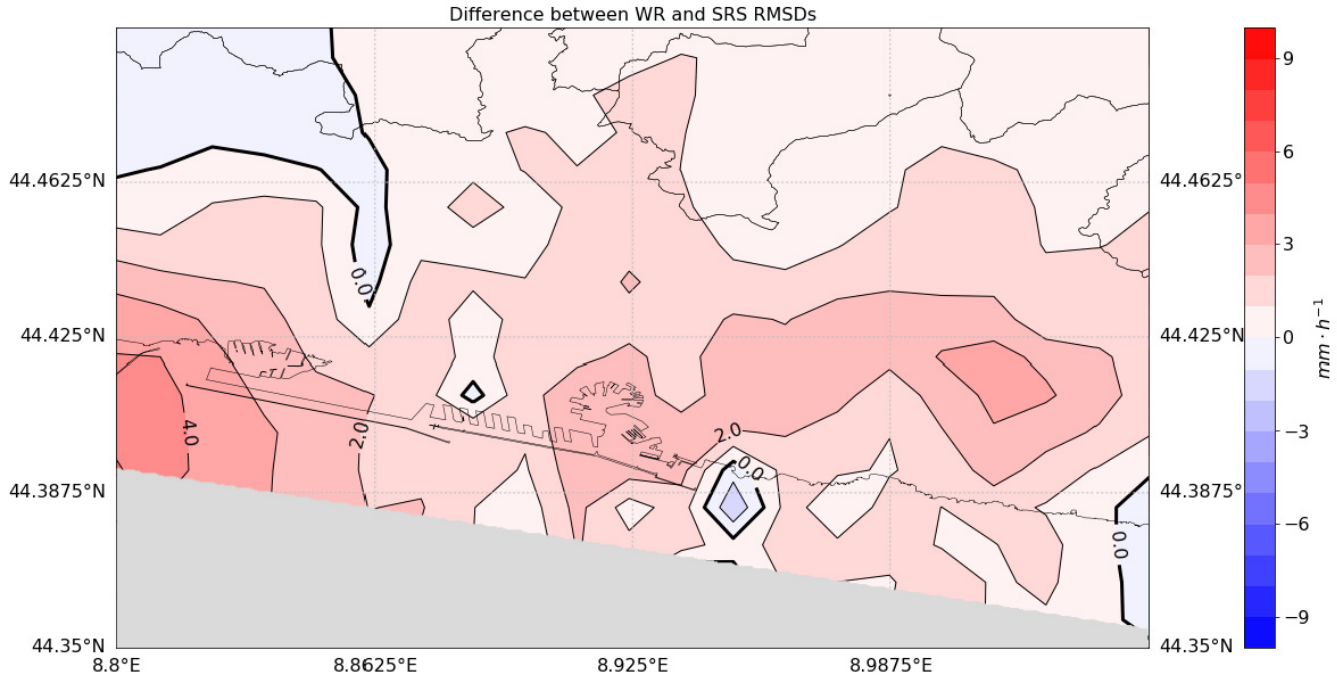


Fig. 9. Spatial distribution of the differences between the RMSE of the 10-min RI measurements made by WR and SRS during the experimental campaign. The red area is associated with higher RMSE values based on the WR measurements while the blue area represents higher RMSE yielded by the SRS measurements. The offshore area is represented in gray.

TABLE X

ABSOLUTE AND NORMALIZED RMSE (RMSE AND RRMSE) OF THE TOTAL RAINFALL AMOUNT MEASURED BY SRS AND WR (h_{SRS} AND h_{WR}) AT THE TBRG LOCATIONS (SEE TABLE V)

Station ID	RMSE(h_{SRS})	RMSE(h_{WR})	rRMSE(h_{SRS})	rRMSE(h_{WR})
	mm	*mm*	-	-
GECTR	52.2	24.3	2.41	1.12
MGAZZ	12.2	15.3	0.65	0.82
GEPTX	16.0	10.9	0.84	0.57
GEQUE	12.4	19.1	0.57	0.88
GEPCA	11.7	18.6	0.53	0.83
PREMA	10.5	31.4	0.33	0.97
RIGHI	18.0	13.9	0.93	0.72
Mean	19.0	19.1	0.89	0.84

TABLE XI

ROOT MEAN SQUARE OF THE RELATIVE ERRORS OF RI MEASURED BY SRS AND WR AT THE TBRG LOCATIONS (SEE TABLE V)

Station ID	RMSE(e_{SRS})	RMSE(e_{WR})
	-	-
GECTR	2.9	1.0
MGAZZ	1.7	1.4
GEPTX	1.4	1.4
GEQUE	1.5	1.3
GEPCA	1.1	1.2
PREMA	1.2	1.0
RIGHI	1.5	1.4
Mean	1.6	1.3

is needed in interpreting the comparison between WR and SRS for the October 27, 2018, event due to a temporary radar calibration problem during the first phase of the event, which was detected *a posteriori* and could not be corrected.

On the other hand, the rainfall cumulated at the event scale showed a higher random behavior of the SRS measurements with respect to WR. This is well highlighted in Fig. 8 by the coexistence of negligible and significant differences $h_{\mathrm{SRS}} - h_{\mathrm{TBRG}}$ at different locations during the same rainfall event. The same issue is less relevant when WR measurements are considered. In the latter case, significant $h_{\mathrm{WR}} - h_{\mathrm{TBRG}}$ figures are visible but their distribution is more uniform among the different measuring locations during a given precipitation event. Table X confirms that the overall behavior of SRS in measuring the total cumulated rainfall is characterized by slightly higher uncertainties than the WR system. Averaging the rRMSE of h among the different TBRG locations yields values equal to 0.89 in the case of SRS and 0.84 when WR maps are considered.

In order to provide a spatial information of the performance of the SRS and WR products over the area of investigation, Fig. 9 shows the distribution of the difference between the RMSE of 10-min $\mathrm{RI_{SRS}}$ and $\mathrm{RI_{TBRG}}$ measured in the City of Genoa during the whole two-year period of analysis.

It is shown that a large portion of the territory is characterized by positive values of the RMSE difference (red color), hence associated with a higher agreement between SRS and TBRG, fields while the negative values (blue shaded areas) are associated with a better agreement between WR and

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TBRG fields. A comparison with the SRS map of Fig. 2 reveals that negative values of the RMSE difference are present in peripheral areas where the SRS microwave down-links are not present or in front of the coastline where a stronger density of down-links may have a negative impact on the agreement of the reconstructed SRS precipitation fields with respect to the TBRG measurements. The last observation is explained by the fact that Fig. 4 showed a higher variability of the SRS performance (less precision) with respect to WR and is also confirmed by Table XI that indicates a higher RMSE of the 10-min RI_{SRS} relative error in these regions (particularly close to GEPCA and in the west of MGAZZ).

It is worth noting that above the sea SRS can measure precipitation in inshore zone only, close to the coastline. In addition, the extrapolation of the TBRG reference measurements is not possible off the coast. For these reasons, the comparison in the offshore zone of Fig. 9 has not been computed and is represented by a gray area.

V. Conclusion

The field experimental activity undertaken in this article supports the results of recent research initiatives by Mercier *et al.* [21], Indrayani *et al.* [17], and Colli *et al.* [7] (see [16], [20]), where satellite microwave down-links were reported to successfully measure RI. Such research efforts were characterized by the execution of comparative analysis between the path-averaged measurements individually provided by SML sensors and ground-based rain gauges.

This article addresses the need for evaluating the applicability of multiple SML measurements to reconstruct RI fields at short time intervals for real-time applications. To this purpose, a network of SRS SML sensors has been set-up at the urban scale in the City of Genoa (Italy) and the measurements were compared with data sets produced by the regional TBRG network and the WR covering over two years of observations. A general conclusion from this article conducted here is a confirmation of the point that SML can be conveniently exploited as a complementary monitoring system to provide spatially distributed information about the space-time evolution of rainfall for real-time applications.

More specific observations derived from this article are summarized as follows.

1) The contingency matrixes of RI showed that the level of agreement between RI values provided by SRS maps and ground-based TBRG measurements is comparable with the results achieved by WR products (Tables V and VI).
2) The statistical distribution of the relative error e of 10-min RI measurements showed that the WR measurements have a slightly better precision (represented by the width of the e distribution and its RMSE) than SRS while the trueness (represented by the mean magnitude of e) provides better figures for SRS.
3) The RMSE of the peak of RI assumed lower values when the SRS data set is analyzed, demonstrating a better behavior of SRS with respect to WR in measuring high-intensity cases.
4) On the other hand, the SRS-cumulated rainfall showed more random performance with the coexistence of negligible and significant differences with respect to the TBRG observations at different locations during the same rainfall event.
5) The spatial distribution of the RMSE of 10-min RI computed over the City of Genoa and considering the whole two-year data set generally showed a stronger agreement between SRS and TBRG maps with respect to WR and TBRG maps.

Notwithstanding the positive indications provided by this article, several improvements should be undertaken to improve the SRS rainfall retrieval methodology. In the following list, we provide an overview of the major follow-ups in order of decreasing priority.

1) Assessing the level of uncertainty introduced by the estimation method of the melting layer altitude. In this article, the environmental temperature measurements performed at different altitudes by two weather stations were used to extrapolate the zero-degree level. Further analysis can be performed by taking advantage of additional information on the vertical profile of environmental temperature such as those provided by nonhydrostatic atmospheric models.
2) Assessing the SRS measurements' uncertainty related to the assumption of a constant spatial distribution of the precipitation rate along the microwave link line.
3) Testing the advantage of introducing more sophisticated map reconstruction techniques, such as those proposed in [8], [14], and [19], in the SRS data processing framework.
4) Investigating the effect of type of precipitation (e.g., convective versus stratiform and orographic) and drop size distribution on the down-link signal attenuation by means of field observations made by disdrometers. Such investigation would help understanding the sensitivity of the ITU-R P838 [17] parameters with respect to the microphysical features of the precipitation and any possible advantage of analyzing the down-link signals available in both the horizontal and the vertical polarizations.
5) Accounting for the bright band effects in the propagation of the Ku frequency electromagnetic waves following a proper approach such as those proposed by Gray *et al.* [16].

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